Training a small network

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Training a small network before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Updating parameters

Without opening the old lesson, explain one key idea from Updating parameters. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Slopes and local change

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Forward passes by hand

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Use one hidden ReLU unit h = ReLU(x) and an **output** y = vh. The hidden **input** weight is fixed at 1, its bias is fixed at 0, and the **output** bias is fixed at 0. Only **output** weight v is trained.

This is a restricted two-layer network, not **training** every parameter of a general network.

Training pairs are (x, target) = (1, 2) and (2, 4). Both x values are positive, so hidden activations are 1 and 2. Define ordinary MSE over both rows.

Start at v = 0 and use learning rate 0.1 for exactly four full-data updates. The declared budget prevents stopping only when a preferred result appears.

Update count	v	Training MSE
0	0	10
1	1	2.5
2	1.5	0.625
3	1.75	0.15625
4	1.875	0.0390625

Each row evaluates the parameter value after that many updates. Row zero records the starting model. The loss decreases in this specific experiment; that pattern is not a promise for all networks or rates.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Training a small network trace and record the first step you can explain without help.

Use W2: Check the mean convention for **Training** a small network. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Training a small network explanation against the supplied facts before discussing it.

Use the worked case for **Training** a small network. Proposed explanation: “Every parameter was trained”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Run a new training case independently

Keep h = ReLU(x), train only v, and use one pair (2, 2), v = 0 and rate 0.125. Use squared error and gradient 2 times (2v - 2) times 2. Perform two updates, reporting each new v and loss.

Explain why the second update changes nothing.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Trace the first update

> Trace the first update. Apply the supplied Training a small network facts and keep your reasoning.

At v = 0, calculate both **prediction**s, errors, MSE and gradient, then update at rate 0.1.

> **My reasoning and evidence**

WORKBOOK · 6 OF 15

W2 Check the mean convention

> Check the mean convention. Apply the supplied Training a small network facts and keep your reasoning.

At v = 1, the sum of squared errors is 5. A learner uses gradient -10 with rate 0.1. Explain the mismatch with this lesson and give the next v under MSE.

My reasoning and evidence

W3 Freeze before checking

> Freeze before checking. Apply the supplied Training a small network facts and keep your reasoning.

At final v = 1.875, calculate the **prediction** and squared error on reserved pair (3, 6). May its target be included in the four reported **training** updates?

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Run a new training case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Run a new training case independently. Apply the supplied Training a small network facts and keep your reasoning.

Keep h = ReLU(x), train only v, and use one pair (2, 2), v = 0 and rate 0.125. Use squared error and gradient 2 times (2v - 2) times 2. Perform two updates, reporting each new v and loss.

Explain why the second update changes nothing.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Expose a frozen representation limit

> Expose a frozen representation limit. Apply the supplied Training a small network facts and keep your reasoning.

For the lesson architecture at x = -1, give h and **prediction** for any finite v. If target is -2, calculate squared error and explain why changing v cannot fix it.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Training a small network to Image-pattern Network Investigation, identify a limit and choose the next useful test.

Add a labelled **training** a small network **evidence** sheet to Image-pattern Network Investigation. Show how you can trace repeated forward loss gradient and update steps.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Training a small network

> Train a single output weight while clearly marking the frozen hidden computation. Explain the labelled step before checking the changed condition.

Train a single **output** weight while clearly marking the frozen hidden computation.

Toy h=ReLU(x), y=vh. One **training** pair x = 3, target 3; start v = 0; squared loss; rate 1/18.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/12

Training a small network

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Freeze hidden step

h=3 for this pair; only v is trainable.



2. Compute initial gradient

Loss 9; gradient 2(3v-3) x 3=-18 at v=0.



3. Update and verify

v becomes 1; prediction 3 and loss 0. Next gradient is 0.

CHANGE ONE THING

Change target from 3 to6 and restart v = 0 with the same rate.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which parameter changed, and which part was not trained? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Image-pattern Network Investigation record.

Project: Image-pattern Network Investigation

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Train a single **output** weight while clearly marking the frozen hidden computation.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? v becomes 1; **prediction** 3 and loss 0. Next gradient is 0.

1. Freeze hidden step

2. Update and verify

3. Compute initial gradient

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Update and verify v becomes 1; **prediction** 3 and loss 0. Next gradient is 0. Only the **output** weight v is being fitted in this toy network; the hidden ReLU is fixed.

The **training input** gives hidden value three. Squared-error slope points v from zero towards one, and this specially chosen rate reaches one in a single update.

Zero loss on one **training** pair is exact **fitting** of that pair, not **evidence** about other **input**s.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change target from 3 to6 and restart v = 0 with the same rate. Which conclusion is supported by the supplied facts?

1. First gradient=-36; v becomes 2; **prediction** 6 and loss 0.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: First gradient=-36; v becomes 2; **prediction** 6 and loss 0. The hidden value remains 3, but the target residual and required **output** weight double.

Repair a missing building block

Review Updating parameters (ai-5/11). Goal: Apply a learning rate to an ordered gradient. Short practice: Why must new loss be recomputed rather than assumed smaller?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Slopes and local change (ai-5/9). Goal: Calculate an average rate of change over a stated interval. Short practice: Which calculation disproves “any increase helps”?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Forward passes by hand (ai-5/6). Goal: Record scores and activations in dependency order. Short practice: Why should feedback identify h before correcting only y?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

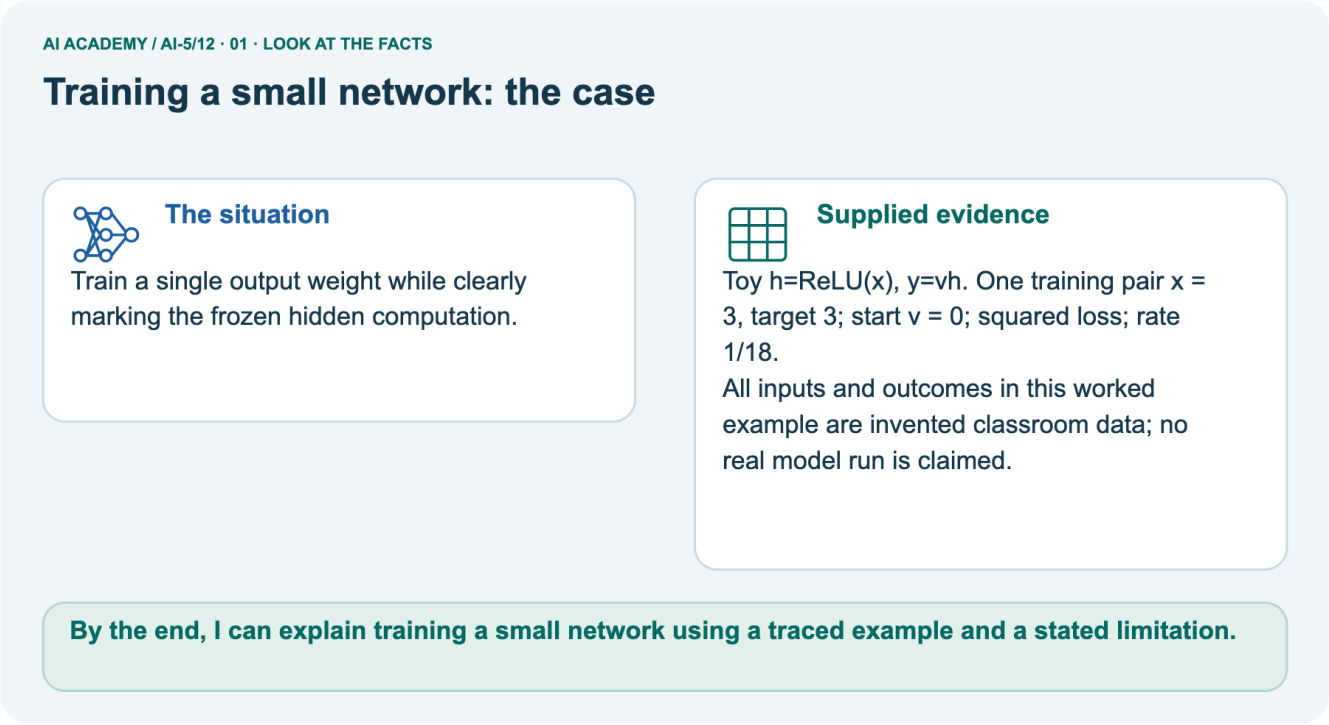
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Training a small network picture from its labels, then record a relevant limit.

01. Training a small network: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

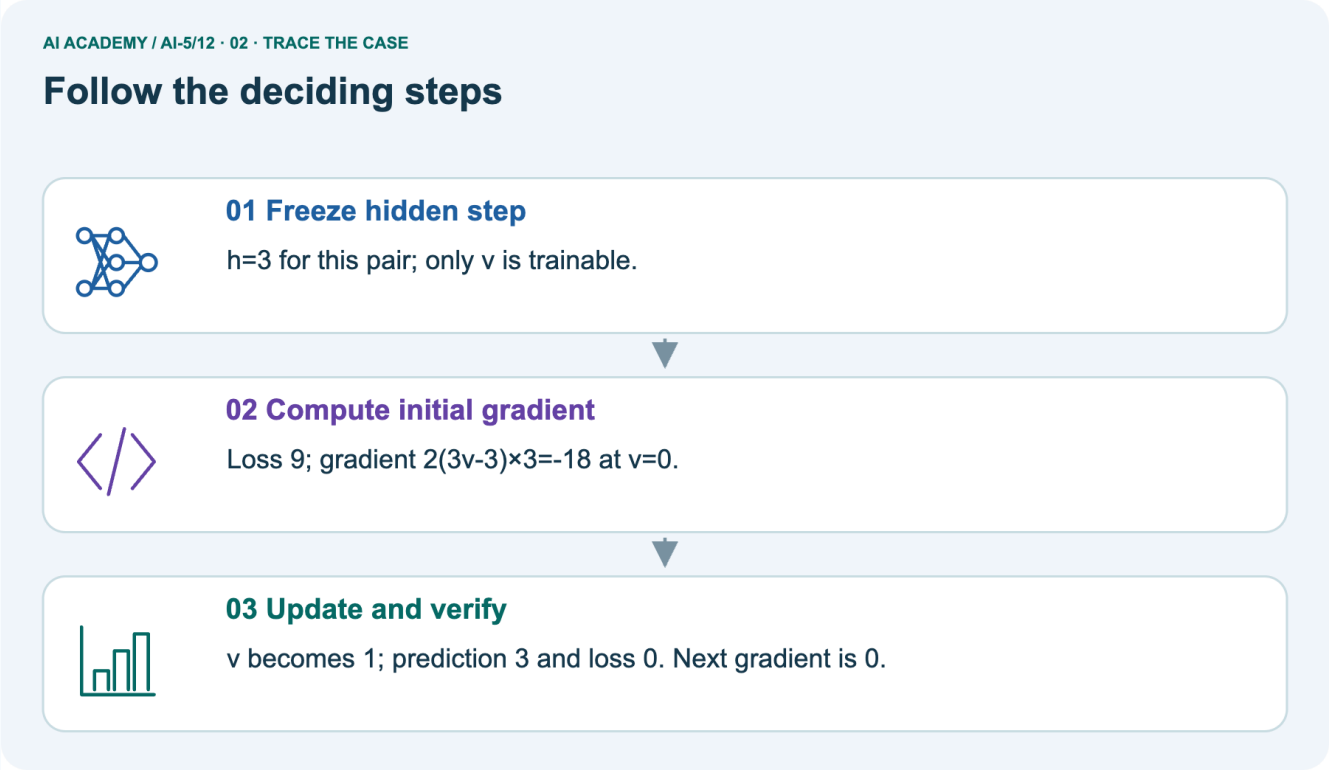
Train a single **output** weight while clearly marking the frozen hidden computation. Supplied facts: Toy h=ReLU(x), y=vh. One **training** pair x = 3, target 3; start v = 0; squared loss; rate 1/18.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze hidden step: h=3 for this pair; only v is trainable. → Compute initial gradient: Loss 9; gradient 2(3v-3)×3=-18 at v=0. → Update and verify: v becomes 1; **prediction** 3 and loss 0. Next gradient is 0.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. One full-data update from v is 1

AI ACADEMY / AI-5/12 · 03 · INSPECT THE LESSON MODEL

One full-data update from v is 1

Stage	Compute with current v	Result
Forward and loss	Predictions [1, 2] for targets [2, 4]	Errors [-1, -2], MSE 2.5
Gradient	2 times (-1 times 1 - 2 times 2) / 2	Gradient -5
Update with rate 0.1	1 - 0.1 times -5	Next v is 1.5

Each row depends on the preceding calculation. Recompute predictions and loss on the next iteration; do not keep using the first gradient after parameters change.

Follow the labelled picture with the matching explanation. Open the vector diagram

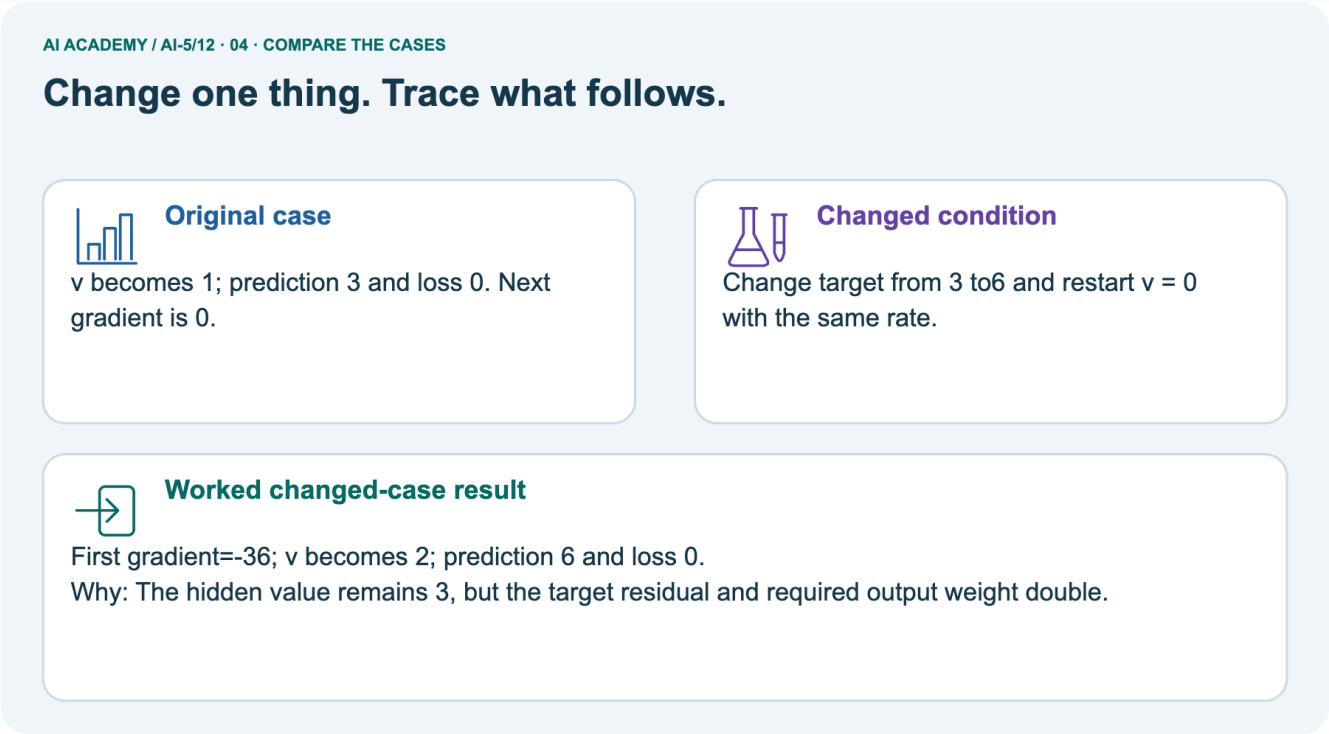
One full-data update from v is 1. Forward and loss | **Prediction**s [1, 2] for targets [2, 4] | Errors [-1, -2], MSE 2.5. Gradient | 2 times (-1 times 1 - 2 times 2) / 2 | Gradient -5.

Update with rate 0.1 | 1 - 0.1 times -5 | Next v is 1.5

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

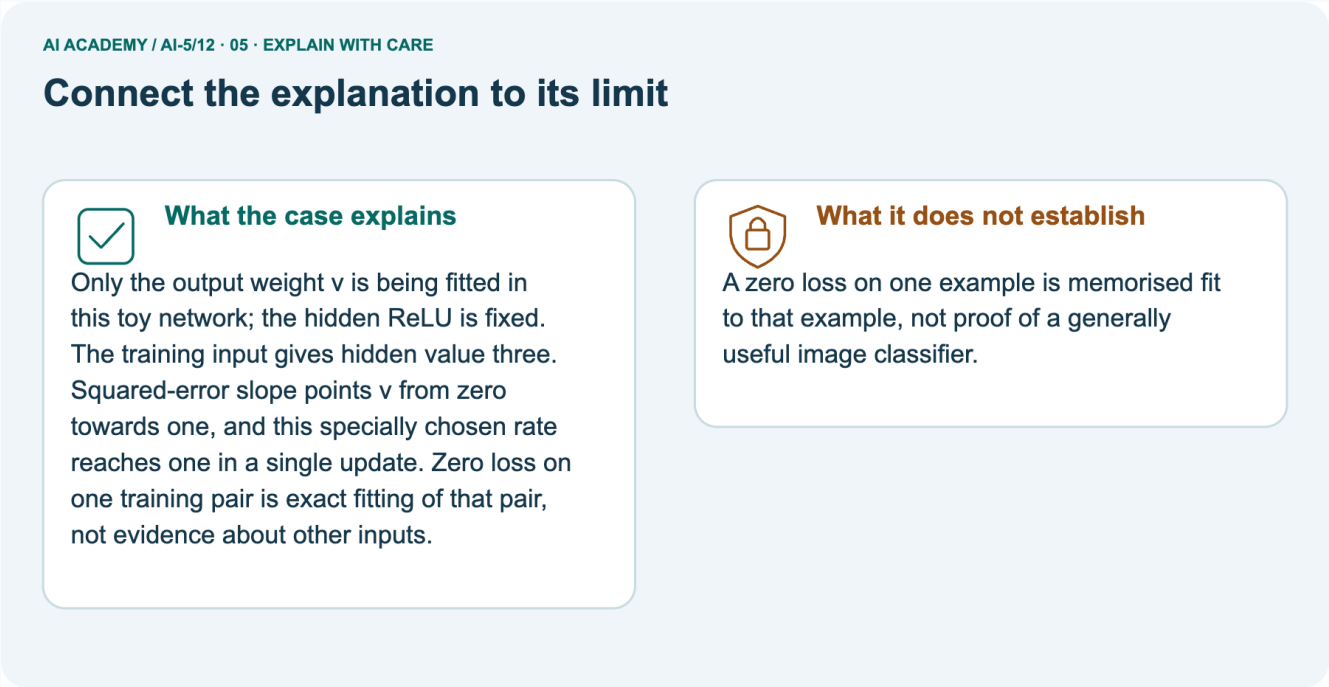
Original case: v becomes 1; **prediction** 3 and loss 0. Next gradient is 0. Changed condition: Change target from 3 to6 and restart v = 0 with the same rate.

Expected result: First gradient=-36; v becomes 2; **prediction** 6 and loss 0. Reason: The hidden value remains 3, but the target residual and required **output** weight double.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Only the **output** weight v is being fitted in this toy network; the hidden ReLU is fixed. The **training input** gives hidden value three.

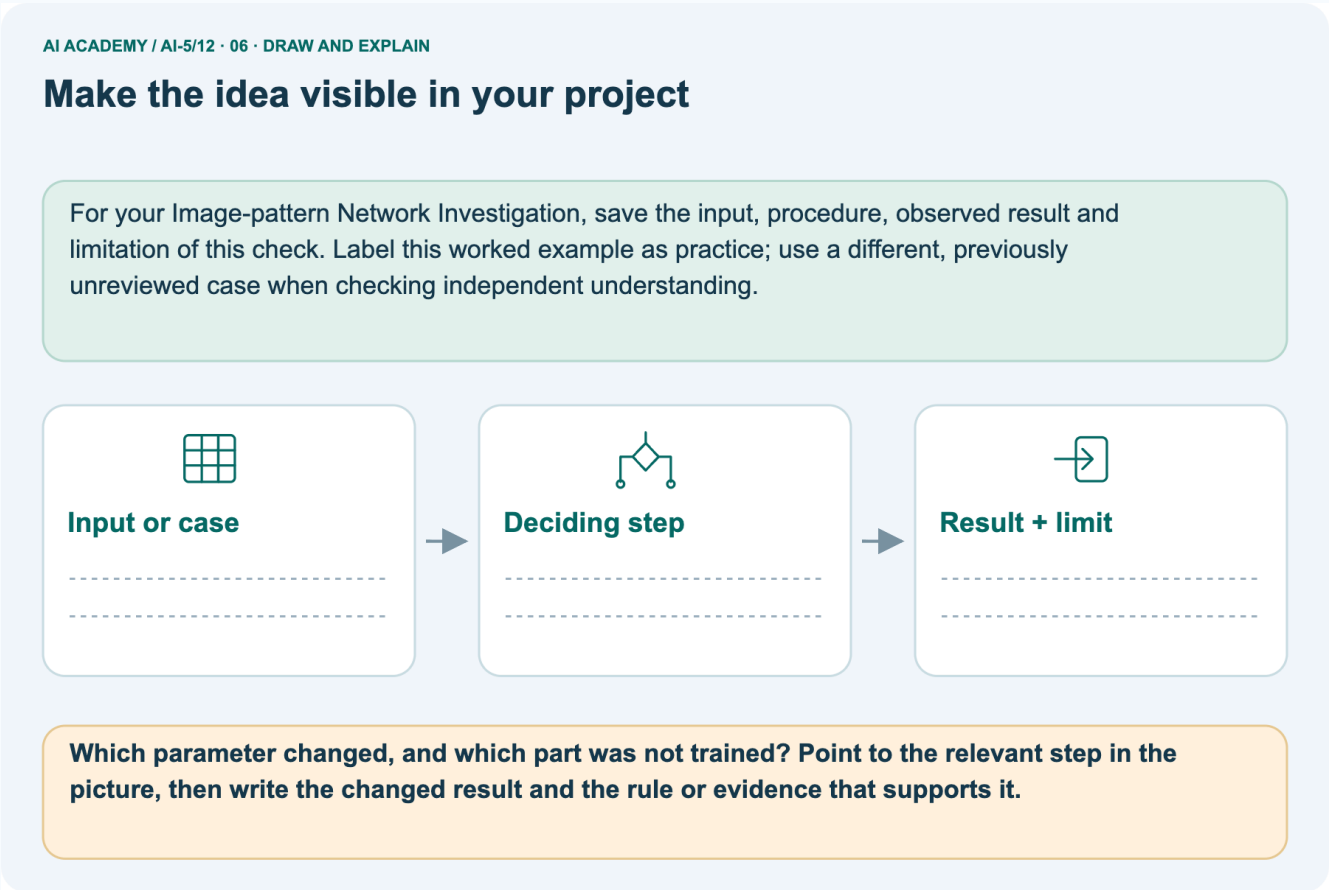
Squared-error slope points v from zero towards one, and this specially chosen rate reaches one in a single update. Zero loss on one **training** pair is exact **fitting** of that pair, not **evidence** about other **input**s.

Limit: A zero loss on one example is memorised fit to that example, not proof of a generally useful image classifier.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which parameter changed, and which part was not trained? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.